

Table 1: POPC and PoxnoPC composition at each oxidation level. The last column shows the percentage of oxidized lipids relative to all PO-type glycerophospholipids (POPC + POPE + POPI + POPS = 85%).

Oxidation level (%)	POPC (%)	PoxnoPC (%)	Oxidized PO lipids (%)
0	47.0	0.0	0.0
25	35.2	11.8	13.9
50	23.5	23.5	27.6
75	11.8	35.2	41.4
100	0.0	47.0	55.3

Table 2: Area per lipid (APL, Å²) of individual lipid species as a function of oxidation level. Values were calculated over the final 50 ns of each trajectory, averaged over all analyzed frames for each replica. Final values represent mean ± standard deviation from three independent replicas.

	POPC	PoxnoPC	POPE	CHL	POPI	POPS
0%	55.7 ± 0.2		50.5 ± 0.8	38.2 ± 0.4	55.2 ± 0.3	53 ± 1
25%	57.7 ± 0.6	59 ± 1	52.2 ± 0.3	38.2 ± 0.6	58 ± 1	55 ± 1
50%	58.2 ± 0.9	61.4 ± 0.2	51.3 ± 0.6	41.5 ± 0.8	57.7 ± 0.5	54.4 ± 0.5
75%	56.2 ± 0.5	61.6 ± 0.6	52 ± 1	40 ± 1	57.2 ± 0.5	54.8 ± 0.7
100%		61.8 ± 0.2	54.1 ± 0.8	40.1 ± 0.4	57 ± 1	58 ± 1

Table 3: Percentage distribution of POPC and PoxnoPC molecules participating in proto-clusters (3 – 4 lipids), small clusters (5 – 10 lipids), and large clusters (> 10 lipids) at different POPC oxidation levels. Values represent mean ± standard deviation from three independent replicas.

	POPC+PoxnoPC		
	proto cluster	small cluster	large cluster
0%	23.3 ± 0.7	8.4 ± 0.7	0.12 ± 0.05
25%	22.5 ± 0.5	8.4 ± 0.7	0.06 ± 0.03
50%	20.9 ± 0.7	9.0 ± 0.8	0.4 ± 0.1
75%	17.9 ± 0.8	5.0 ± 0.4	0.02 ± 0.02
100%	19.3 ± 0.7	6.3 ± 0.3	0.06 ± 0.03

Table 4: Percentage distribution of POPC, PoxnoPC, and CHL molecules participating in proto-clusters, small clusters, and large clusters at different POPC oxidation levels. Values represent mean ± standard deviation from three replicas.

	POPC+PoxnoPC+CHL		
	proto cluster	small cluster	large cluster
0%	14.2 ± 0.5	28.0 ± 0.8	39.7 ± 0.7
25%	15.1 ± 1.5	27.9 ± 1.2	37.8 ± 1.3
50%	15.1 ± 1.5	25.3 ± 0.9	35.4 ± 1.4
75%	17.9 ± 0.9	30.7 ± 1.4	24.2 ± 1.1
100%	15.9 ± 0.7	31.1 ± 1.0	25.9 ± 0.7